

Normalization

- ① 1NF (1st Normal Form): Multivalued এর আলাদা Table
- ② 2NF: Partial Dependency থাকবে না
- ③ 3NF: Transitive Dependency থাকবে না

<u>Sid</u>	sName	Dept-ID	Dept-Name
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student ID $\rightarrow$ Dept ID

Dept ID $\rightarrow$ DName

Sid $\rightarrow$ DName

Normalization Example

Student

<u>StID</u>	<u>Course Code</u>	<u>Semester</u>	Student Name	Attempted Credits	Course Title	Grade	Proj ID	Project Marks	Project Name
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Additional Functional Dependencies

- $\rightarrow$ FD1: StudentID $\rightarrow$ StudentName
- $\rightarrow$ FD2: CourseCode $\rightarrow$ CourseTitle
- $\rightarrow$ FD3: StudentID, Semester $\rightarrow$ AttemptedCredits
- FD4: ProjID $\rightarrow$ ProjectName, ProjectMarks

Question: Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete "current" schema after each normalization step.

Already 1NF

2NF:

student (Sid, sName)

CourseCode (Ccode, Ctitle)

Credit (Sid, Semester, ACredit)

Grades (Sid, Ccode, Semester, Grade, Pid, Pmark, Pn)

3NF:

student (Sid, sName)

Course (Ccode, CTitle)

Credit (Sid, Semester, ACredit)

Grade (Sid, Ccode, Semester, Grade, Pid)

Project (Pid, Pname, Pmark)

Question 3:

Consider the following relational Schema:

Books (AuthorName, BookTitle, Publisher, Price, Year, AuthorAffiliation, Type, Category)

The primary key is underlined.

The schema has the following additional Functional Dependencies:

FD1 AuthorName → AuthorAffiliation

FD2 BookTitle → Publisher, Price, Year, Type, Category

FD3 Type → Category

Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete “current” schema after each normalization step.

1NF

2NF



Answer 3:

It is in 1NF because no multivalued/composite attributes or nested relations present in the given schema. But, not in 2nf as partial dependency on the primary key present due to the first two FDs and the third FD causes transitive dependency so also not in 3NF.

Normalizing to 2NF:

Author(AuthorName, AuthorAffiliation)

Book1 (BookTitle, Publisher, Price, Year, Type, Category)

Book2 (AuthorName, BookTitle)

Normalizing to 3NF:

Author(AuthorName, AuthorAffiliation)

Book1 (BookTitle, Publisher, Price, Year, Type)

BookType(Type, Category)

Book2 (AuthorName, BookTitle)



Question 2:

Consider the following relational Schema:

Books (ISBN, AuthorName, BookTitle, Publisher, Price, Year, AuthorAffiliation, Type, Category)

The primary key is underlined.

The schema has the following additional Functional Dependencies:

FD1 AuthorName → AuthorAffiliation

FD2 Type → Category

Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete “current” schema after each normalization step.

1NF

2NF

Answer 2:

It is in 2NF. 2nf has 2 conditions: 1. should be in 1NF and 2. Should not have any partial dependency on the primary key. It is already in 1NF because no multivalued/composite attributes or nested relations present in the given schema and also there are no partial dependencies. So it is currently in 2NF.

It is not in 3nf because there are transitive dependencies due to the additional FDs.

Normalizing to 3nf:

Books (ISBN, AuthorName, BookTitle, Publisher, Price, Year, Type)

Author(AuthorName, AuthorAffiliation)

BookType (Type, Category)

